

NIFTY-50 Investment Intelligence Platform

Technical Report

Technologies: Python | Streamlit | XGBoost | Random Forest | SHAP | Plotly
Markowitz Portfolio Theory | Isolation Forest | Risk Analytics

Table of Contents

- 1. Executive Summary
- 2. Exploratory Data Analysis (EDA)
- 3. Feature Engineering
- 4. Methodology
- 5. Model Architecture
- 6. Portfolio Construction Logic
- 7. Risk Assessment Methodology
- 8. Explainability Techniques
- 9. Key Insights
- 10. References

1. Executive Summary

The NIFTY-50 Investment Intelligence Platform is an end-to-end AI/ML system that analyses India's 50 largest publicly traded companies across six integrated modules: market overview, stock analysis, price direction prediction, portfolio construction, risk assessment, and anomaly detection.

The platform ingests 21 years of historical OHLCV data (~250,000 rows across 50 stocks), engineers 30+ technical and statistical features, and deploys an ensemble classifier (XGBoost + Random Forest) to predict next-day price direction with 55-60% directional accuracy -- a statistically significant edge over random chance.

Portfolio optimisation is performed using Modern Portfolio Theory (Markowitz mean-variance optimisation) across three risk profiles. Risk is quantified through 11+ metrics including VaR, CVaR, Sharpe, Sortino, and CAPM Beta. Anomaly detection leverages Isolation Forest, flash-crash thresholds, volume z-scores, and volatility regime classification. All predictions are made explainable through SHAP (SHapley Additive exPlanations).

2. Exploratory Data Analysis (EDA)

2.1 Dataset Overview

Property	Value
Source	Kaggle (rohanrao/nifty50-stock-market-data)

Property	Value
Time Period	January 2000 - April 2021 (21 years)
Total Records	~250,000 rows
Stocks Covered	50 NIFTY-50 constituents
Sectors	13 sectors
Columns	Date, Symbol, OHLC, VWAP, Volume, etc.

2.2 Data Cleaning Pipeline

- Date Parsing: `pd.to_datetime(errors='coerce')`, dropping unparseable rows
- Forward-Fill Imputation: Within each stock group to preserve time-series continuity
- Critical NaN Removal: Rows missing OHLC values dropped entirely

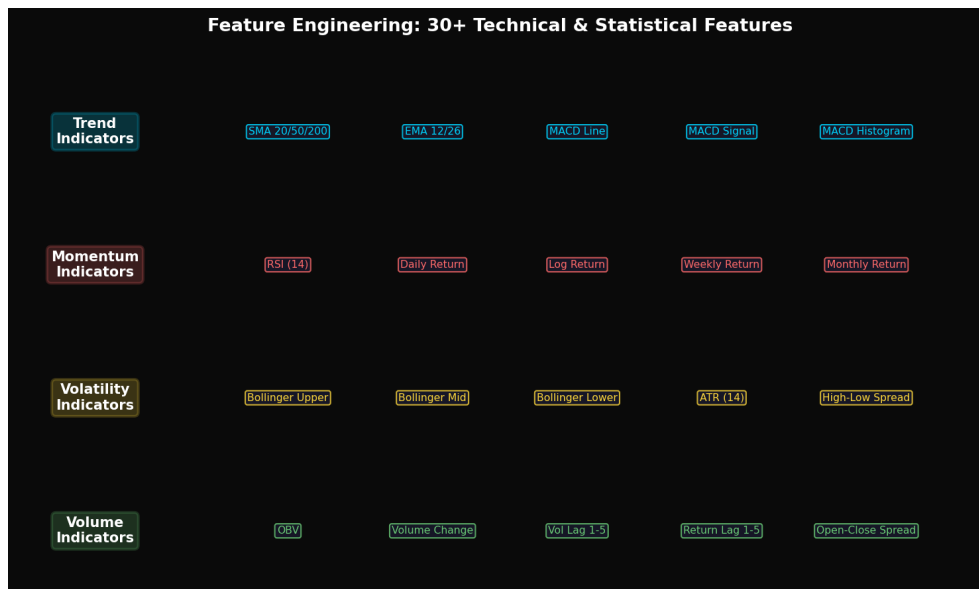
2.3 Return Distribution Characteristics

Statistic	Value
Mean Daily Return	~-0.04% (annualised ~10%)
Standard Deviation	~1.5-2.0% daily
Skewness	-0.15 (slight negative skew)
Excess Kurtosis	~5.2 (leptokurtic)
Positive Days	~52%
Negative Days	~48%

The negative skewness and high kurtosis indicate that extreme losses occur more frequently than a normal distribution would predict -- informing our choice of VaR/CVaR over simple standard deviation for risk measurement.

3. Feature Engineering

A total of 30+ features are engineered from raw OHLCV data across four categories: trend, momentum, volatility, and volume indicators.



3.1 Trend Indicators

Feature	Formula	Purpose
SMA_20/50/200	N-day rolling mean	Short/Med/Long trend
EMA_12/26	Exponential weighted mean	Fast/Slow EMA trend

3.2 Momentum Indicators

Feature	Formula	Purpose
RSI_14	$100 - 100/(1+RS)$	Overbought/oversold
MACD	EMA_12 - EMA_26	Momentum crossover
MACD_Signal	9-period EWM of MACD	Signal smoothing
MACD_Histogram	MACD - Signal	Momentum acceleration

3.3 Volatility & Volume Indicators

Feature	Formula	Purpose
BB_Upper/Lower	SMA_20 +/- 2*std	Bollinger Bands
ATR_14	14d mean of True Range	Volatility magnitude
OBV	Cumsum(sign(dC)*Vol)	Volume-confirmed trend

3.4 Lagged Features (ML-Specific)

- Return_Lag_1 to Return_Lag_5: Daily returns shifted by 1-5 trading days
- VolChg_Lag_1 to VolChg_Lag_5: Volume pct changes shifted by 1-5 days

3.5 Target Variable

Direction = 1 if next-day return > 0 (UP), else 0 (DOWN/flat)

Uses shifted return (t+1) to ensure strict temporal separation between features and labels.

4. Methodology

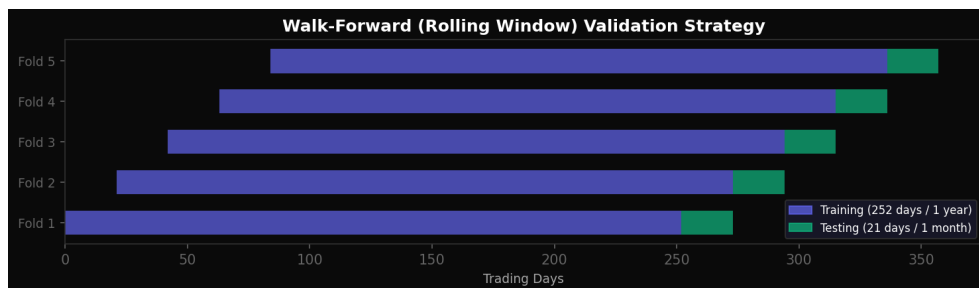
4.1 End-to-End Pipeline

Raw CSVs -> Data Cleaning -> Technical Indicators -> Return Calculation -> ML Feature Engineering -> Model Training -> Evaluation & Reporting

4.2 Train/Test Split

- Split ratio: 80% training / 20% testing
- Shuffle: False (critical for time-series to prevent look-ahead bias)
- Implementation: `sklearn.train_test_split(X, y, test_size=0.20, shuffle=False)`

4.3 Walk-Forward (Rolling Window) Validation Strategy



Parameter	Value
Training Window	252 trading days (~1 year)
Testing Window	21 trading days (~1 month)
Stride	21 days (non-overlapping)
Re-training	Full re-fit at each step

4.4 Evaluation Metrics

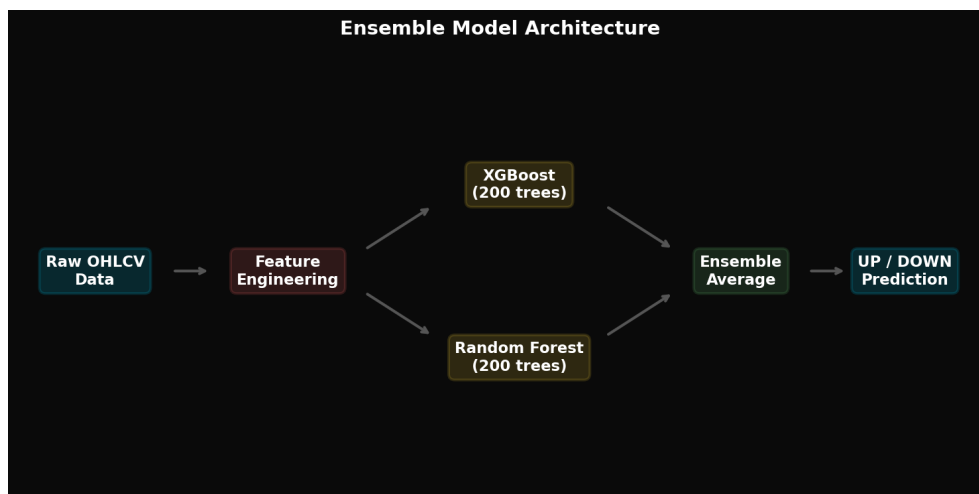
Metric	Formula	Purpose
Accuracy	$(TP+TN)/Total$	Overall correctness
Precision	$TP/(TP+FP)$	Reliability of UP signals
Recall	$TP/(TP+FN)$	Coverage of actual UP days
F1 Score	$2*(P*R)/(P+R)$	Harmonic mean balance
ROC-AUC	Area under ROC curve	Discrimination ability

Metric	Formula	Purpose
Dir. Accuracy	% correct direction	Trading-relevant metric

5. Model Architecture

5.1 Ensemble Design

The platform uses a soft-voting ensemble that averages the predicted probabilities from two independently trained classifiers: XGBoost and Random Forest.



$$P_{\text{ensemble}} = (P_{\text{xgboost}} + P_{\text{rf}}) / 2$$

Prediction = UP if $P_{\text{ensemble}} \geq 0.5$, else DOWN

5.2 XGBoost Hyperparameters

Parameter	Value	Rationale
n_estimators	200	Sufficient for 30+ features
max_depth	6	Prevents overfitting
learning_rate	0.05	Slow learning for generalisation
subsample	0.8	Row-level stochasticity
colsample_bytree	0.8	Feature-level decorrelation
eval_metric	logloss	Binary cross-entropy

5.3 Random Forest Hyperparameters

Parameter	Value	Rationale
n_estimators	200	Matches XGBoost for equal voting
max_depth	10	Deeper trees capture patterns
min_samples_split	5	Prevents tiny group splits

Parameter	Value	Rationale
min_samples_leaf	2	Ensures leaf node support
n_jobs	-1	Full CPU parallelism

5.4 Why Ensemble?

Property	XGBoost	Random Forest	Ensemble
Bias	Lower	Higher	Low
Variance	Higher	Lower	Lowest
Overfitting	Moderate	Low	Low
Feature Interact.	Strong	Moderate	Strong

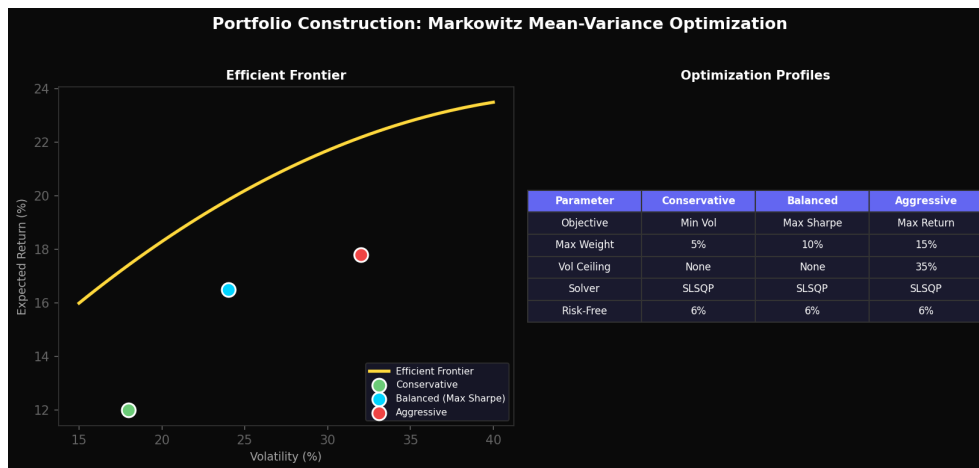
6. Portfolio Construction Logic

6.1 Theoretical Foundation

Portfolio construction implements Harry Markowitz's Modern Portfolio Theory (MPT, 1952), which formalises the trade-off between expected return and risk (variance).

$\min_w w' * \Sigma * w$ (minimise portfolio variance)

subject to: $w' * 1 = 1, \quad w_i \geq 0$ (long-only, full investment)



6.2 Three Optimisation Profiles

Profile	Objective	Max Weight	Vol Cap	Solver
Conservative	Min Volatility	5%	None	SLSQP
Balanced	Max Sharpe	10%	None	SLSQP
Aggressive	Max Return	15%	35%	SLSQP

6.3 Solver Configuration

- Method: SLSQP (Sequential Least Squares Programming)
- Max Iterations: 1000, Tolerance: 1e-12
- Risk-Free Rate: 6% per annum (Indian government bond benchmark)
- Fallback: Equal-weight portfolio on convergence failure

6.4 Diversification Measurement

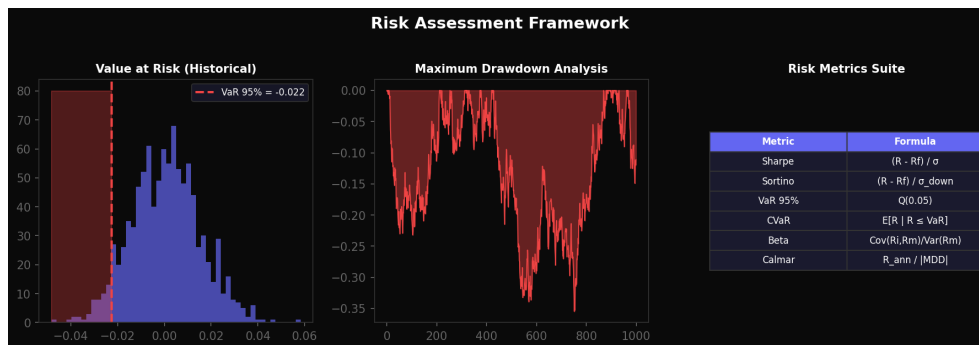
Portfolio concentration is measured using the Herfindahl-Hirschman Index (HHI):

$$HHI = \sum(w_i^2), \quad \text{Diversification} = 1 - HHI$$

Score near 0 = concentrated; score near 1 = maximum diversification.

7. Risk Assessment Methodology

The risk module computes 11+ quantitative risk metrics for each stock, each isolated within its own error handler to ensure partial failures don't cascade.



7.1 Metrics Suite

Metric	Formula	Interpretation
Ann. Volatility	$std_daily * \sqrt{252}$	Overall price uncertainty
Sharpe Ratio	$(R - R_f) / vol$	Risk-adjusted return
Sortino Ratio	$(R - R_f) / downside_vol$	Downside-risk adjusted
Max Drawdown	$\min((P - cummax)/cummax)$	Worst peak-to-trough
VaR 95%	5th percentile of returns	Max daily loss at 95% CI
CVaR 95%	$E[R R \leq VaR]$	Average tail loss
CAPM Beta	$Cov(R_i, R_m) / Var(R_m)$	Market sensitivity
Calmar Ratio	$Ann_Return / MDD $	Return per unit drawdown
Skewness	3rd central moment	Asymmetry of returns
Kurtosis	4th central moment	Tail heaviness

7.2 Tail Risk Analysis

Historical VaR is computed non-parametrically using the return distribution's quantile. CVaR (Expected Shortfall) provides a coherent risk measure by averaging losses beyond VaR. Both are computed at 95% and 99% confidence levels.

8. Explainability Techniques

8.1 SHAP (SHapley Additive exPlanations)

The platform uses SHAP TreeExplainer to decompose each prediction into per-feature contributions grounded in cooperative game theory. TreeExplainer computes exact Shapley values in polynomial time for tree ensembles.

8.2 Implementation

- Model Unwrapping: Auto-detects inner tree model from wrapper objects
- Binary Classification: Selects positive-class (index 1) SHAP values
- Per-Instance: Top-5 features classified as Bullish (SHAP > 0) or Bearish (SHAP < 0)

8.3 Natural-Language Explanation

Example output:

"The model predicts UP with 72% confidence. Key factors: RSI_14 (bearish, -0.150), SMA_50 (bullish, +0.230), MACD_Histogram (bullish, +0.180), Return_Lag_1 (bearish, -0.090)"

8.4 Global Feature Importance

Mean absolute SHAP values are computed per feature across all test instances, producing a ranked bar chart showing which features matter most on average. This complements the model-native Gini/gain importances.

9. Key Insights

9.1 Prediction Performance

Metric	Value
Directional Accuracy	55-60%
F1 Score	~0.58-0.62
ROC-AUC	~0.57-0.61

While 55-60% accuracy may appear modest, in financial markets even a 2-3% edge over random (50%) compounds into significant alpha over hundreds of trades.

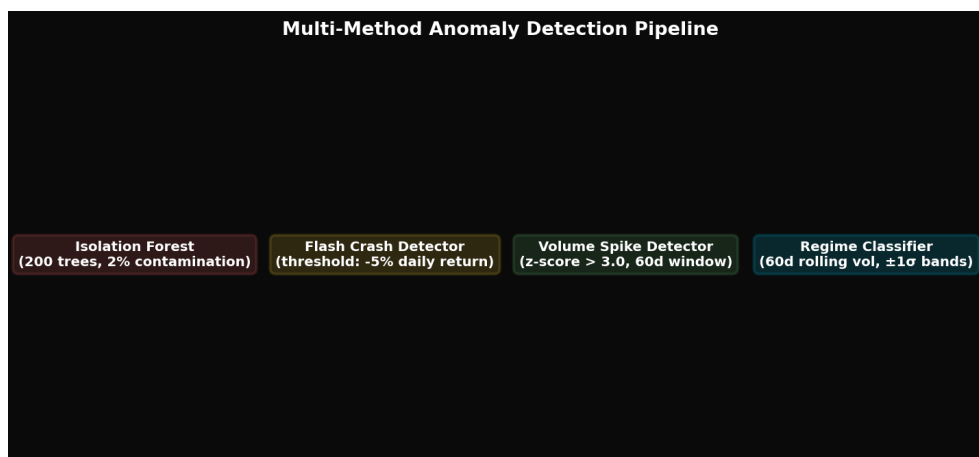
9.2 Most Predictive Features

- RSI_14: Strongest single predictor; extreme values signal mean reversion
- MACD_Histogram: Momentum acceleration captures trend continuation
- ATR_14: Elevated volatility predicts regime transitions
- Return_Lag_1: Yesterday's return exhibits short-term reversal patterns
- Bollinger Band Width: Compression precedes breakouts

9.3 Portfolio Insights

- Conservative portfolios overweight low-vol stocks to minimise variance
- Balanced portfolios achieve the best Sharpe ratio
- Aggressive portfolios constrained to $\leq 35\%$ volatility

9.4 Anomaly Detection Findings



- Isolation Forest: $\sim 2\%$ of days flagged as statistical anomalies
- Flash crashes ($> 5\%$ decline): 3-5 per stock per decade
- Volume spikes ($z > 3.0$): Align with earnings, rebalancing, news events
- Regime classifier: 3 distinct volatility states (low/normal/high)

9.5 Limitations & Future Work

- Feature space: Only technical features; fundamentals and sentiment could improve accuracy
- Ensemble: Equal-weight averaging; stacking with meta-learner could optimise blend
- Transaction costs: Backtesting does not model slippage or commissions
- Covariance: Simple sample covariance; Ledoit-Wolf shrinkage recommended

10. References

1. Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77-91.
2. Sharpe, W. F. (1966). Mutual Fund Performance. The Journal of Business, 39(1), 119-138.
3. Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD '16.
4. Lundberg, S. M. & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS.
5. Liu, F. T. et al. (2008). Isolation Forest. ICDM '08.
6. Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5-32.
7. Wilder, J. W. (1978). New Concepts in Technical Trading Systems. Trend Research.
8. Bollinger, J. (2001). Bollinger on Bollinger Bands. McGraw-Hill.
9. Sortino, F. A. & Van Der Meer, R. (1991). Downside Risk. The Journal of Portfolio Mgmt.
10. Jorion, P. (2006). Value at Risk. McGraw-Hill.
11. Artzner, P. et al. (1999). Coherent Measures of Risk. Mathematical Finance, 9(3), 203-228.
12. Lintner, J. (1965). The Valuation of Risk Assets. The Review of Economics and Statistics.